LAB # 5
To design the vending machine using VHDL and reproduce the results on the FPGA board

Objectives

- Part 1: Design and testing of a complex counter with the help of FSM.
- Part 2: Design and testing of a Vending Machine with the help of FSM.

Pre-Lab

In traditional digital design, we often use Boolean equations to map inputs directly to outputs. However, as systems grow in complexity (like a Vending Machine), the number of input combinations becomes unmanageable.

Instead of designing a massive truth table, we design a **Controller**. An FSM allows us to break a complex problem into "States" (Memory) and "Transitions" (Logic). This lab compares:

1. **Traditional Sequential Logic:** Using counters with simple mode bits.
2. **FSM-Based Control:** Using a central controller to manage complex assets like shared resources or currency.

Part 1: Introduction, specification, design and testing of a Complex Counter

The Multi-Mode "Complex" Counter

In-Lab Task1: Design a 3-bit counter that can switch between **Binary** and **Gray Code** sequences based on a mode input m

Mode (m)	Sequence Type	Pattern
m=0	Binary	0, 1, 2, 3, 4, 5, 6, 7 ...
m=1	Gray Code	0, 1, 3, 2, 6, 7, 5, 4...

While a binary counter is just an adder, a Gray code counter requires non-standard jumps (e.g., 3 → 2 then 2 → 6). By using an FSM, we simply define these jumps as state transitions, making the design significantly easier to verify than trying to derive the logic gates manually.

Part 2: Introduction, specification, design and testing of Vending

Design of sequential circuits is quite a bit more involved than that of combinational circuits. There are many opportunities for things to go wrong, and debugging a malfunctioning circuit can be tedious at best and downright frustrating at worst. Fortunately, by adopting a structured approach to the design and debug of sequential circuits, and by carefully documenting the design, many of these problems become manageable. The analogy to software development here is quite apt; if you write "spaghetti" code you are more likely to be frustrated when things don't work and won't get much sympathy when you seek help. The reward for discipline, on the other hand, is that you'll spend less time chasing after hard-to-find bugs and more enjoying the fruits of your creativity.

Design can be fun!

In this Lab, we will look at making a vending machine. Vending machines are more complicated than you might think. We know that a vending machine must remember how much money has been inserted. This means that its outputs are a function of past inputs, so it must be a sequential circuit. Probably the best way to design this circuit is as a state machine.

Problem Statement:

A soft-drink dispensing machine is one that can dispense from a selection of pre-filled drinks. Depending on the total cash entered in the machine as well as the choice (for the drinks) selected, the machine will put out a can of each type of drink selected.

Dispensing Machine Characteristics:

1. The dispensing machine accepts Rs. 1, Rs. 2, Rs. 4 and Rs. 8 only.
2. It dispenses only three types of drinks:
    i. Rango (costing Rs. 3)
    ii. Next (costing Rs. 4)
    iii. Fizup (costing Rs. 5).
3. The machine has the following 10 inputs:
    i. Clock: You will generate it in test bench and it to your system through port mapping
    ii. Reset: Asynchronous active-
    iii. Rs.1: when Rupee 1 is entered
    iv. Rs.2: when Rupees 2 are entered
    v. Rs.4: when Rupees 4 are entered
    vi. Rs.8: when Rupees 8 are entered
    vii. Rango: when Rango drink is selected
    viii. Next: when Cola drink is selected

ix. Fizup: when Fizup drink is selected

x. ENTER: To trigger the selection

to put out the requested drink as well as the remaining change.

The machine has the following 3 outputs:

i. **Total money deposited:** Using any output port display the value on the wave form in the end of selection of one cold drink, showing the total money deposited (in HEX) at run-time.

ii. **Change:** Another output port will be set to put out the change (in BINARY) for the user when ENTER is asserted. If no drink is selected after money has been entered, then pressing ENTER returns the entire cash back to the customer.

iii. **Dispensed item:** The selected drink/s is/are put out by the machine (depending upon total cash entered and drink choice made) when ENTER is asserted

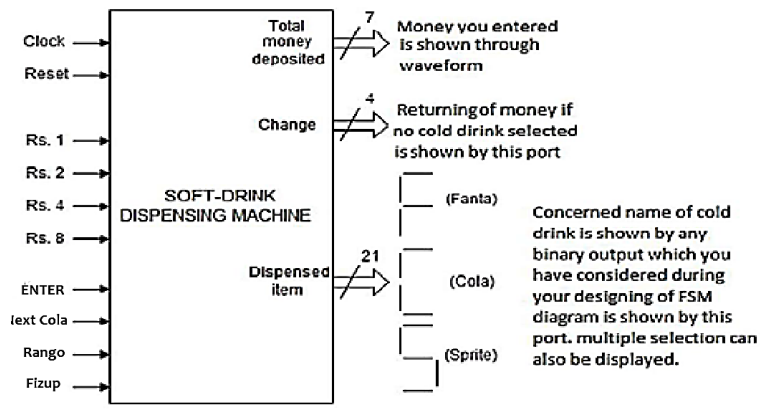

FIGURE 5.1: SOFT DRINK DISPENSING MACHINE

In-Lab Task 2:

Design the vending machine that perform the above-mentioned operations using VHDL language.

The machine should behave in accordance with the following specifications:

1. A customer needs to deposit a sufficient amount of money before or at the same time that he or she is selecting an item for purchase. If the item costs less than the deposited amount, the item is dispensed and the correct change is returned. If the item costs more than the deposited amount, the machine waits for more coins to be inserted or for a cheaper item to be selected.
2. The coin slot in the machine will only accept a single coin. Attempts to jam two or more coins cannot affect the state of the machine and are ignored. Similarly, attempts to make multiple selections (e.g. selecting Rango and Fizup at the same time) are ignored.
3. At any time, pressing the coin return button causes the credit amount to be returned as change in rupees. Furthermore, when credit is equal to Rs 15, any additional coins that are inserted are immediately returned as change.

Rubric for lab assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed all the tasks and showed the results withoutany help of the instructor.	4	
Good	The student completed all the tasks and showed the results with minimal help of the instructor.	3	
Average	The student partially completed the task and showed results.	2	
Worst	The student did not complete the task.	1	

Instructor Signature:_____ Date: _____